



BOARDROOM PRIMER FOR AI ADOPTION & GOVERNANCE

AI MALAYSIA

NATIONAL AI OFFICE

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AI Malaysia Berhad is the central national agency under the Digital Ministry, mandated to lead, coordinate, and accelerate the nation's artificial intelligence (AI) ecosystem towards realising the AI Nation 2030 vision. Operating as the primary catalyst for innovation and national competitiveness, AI Malaysia champions two flagship mandates: executing the National AI Action Plan 2026-2030 to drive cross-sectoral economic growth and productivity, and establishing the Malaysia AI Safety Institute to protect the public through rigorous risk assessments, governance, and technical assurance.

Disclaimer: This Primer is an educational and voluntary resource. It does not constitute national policy, a national guideline, legal, regulatory, accounting, financial or technical advice, nor does it establish a standard of care. It was prepared at a time when Malaysia's AI governance legislation remains under development and may be updated following future legal, regulatory, policy or sectoral developments. AI tools were used to support the drafting and editorial preparation of this Primer. The final content remains subject to human review, and directors and management remain responsible for their own decisions and for ensuring compliance with applicable laws, regulations, listing rules and governance codes.

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Introduction

Artificial Intelligence (AI) is rapidly transforming the global business landscape, presenting both unprecedented opportunities and complex challenges for organisations worldwide. For business leaders in Malaysia, the potential of AI is being recognised as an opportunity to maintain competitiveness, foster innovation, and ensure sustainable growth. However, effective oversight of AI is crucial, as unmanaged risks can lead to significant financial, reputational, and operational consequences.

Board members need not possess deep technical expertise in AI, but they should cultivate a working grasp of its possibilities, challenges, and the responsibilities it creates. In their capacity as stewards of the organisation, directors are entrusted with ensuring that AI adoption aligns with the company's principles, reflects its strategic direction, and serves its long-term purpose. This stewardship also requires them to safeguard the organisation by maintaining vigilant oversight of the risks, ethical dimensions, and potential liabilities that accompany the use of AI.

AI is no longer a peripheral technology. It is a strategic capability that can reshape competitiveness, risk, and trust. In practice, becoming "AI-ready" is not about technical mastery but about setting the tone for responsible adoption through informed challenge and clear governance expectations.

While providing frameworks and highlighting best practices, this Primer is intended as a voluntary resource. Its content is adaptable across various sectors and organisations of different sizes. Ultimately, this resource aims to begin the journey of equipping business leaders with the confidence, foresight, and ethical stewardship necessary to lead effectively in an AI-driven future.

Purpose and Intended Audience

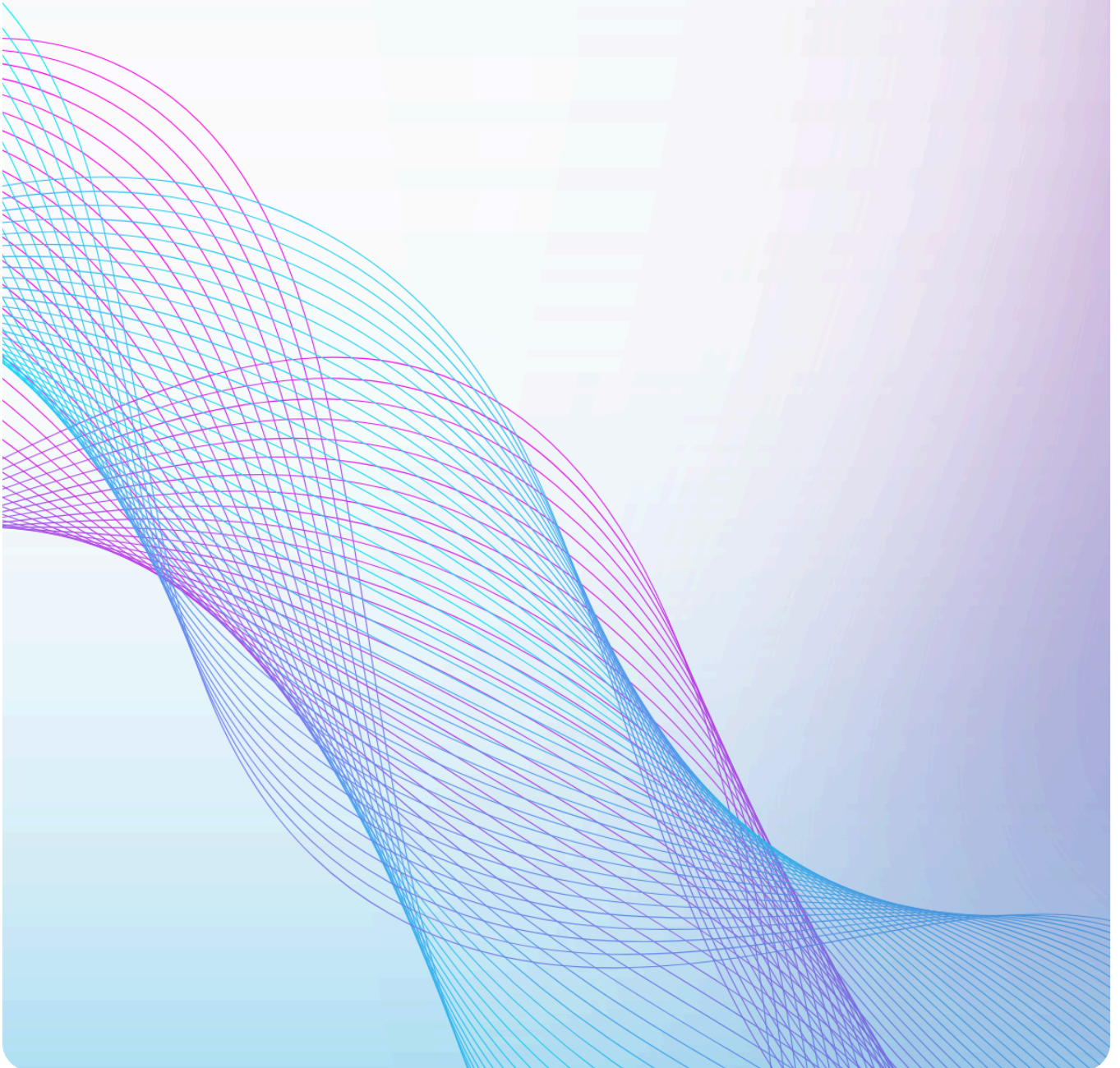
This Primer is designed to support Malaysian boards, directors and senior management in exercising informed oversight of AI adoption and governance. It is not intended to turn board members into technical specialists. Rather, it provides a practical reference to help boards understand AI-related opportunities and risks, ask the right strategic questions and consider whether AI use cases are aligned with organisational purpose, risk appetite, ethical considerations and applicable legal and governance obligations.

The Primer may also be useful for organisations at different stages of AI maturity, from those beginning to explore AI use cases to those deploying AI systems across business functions. In applying this Primer, boards should consider whether AI initiatives create measurable value, whether associated risks are proportionately managed, and whether appropriate governance, validation, monitoring and accountability arrangements are in place before material deployment or scaling.

This Primer is prepared by AI Malaysia (National AI Office). The contribution of the NAIO AI Advisory Working Group convened between February 2025 and June 2025 was valuable to the conceptualisation and development of this document. Additionally, special mention and thanks are given to the Institute of Corporate Directors Malaysia (ICDM), which facilitated feedback sessions on the initial drafts of this document with its members and the working team in NAIO's AI Policy Unit.

This document should be treated as a living document which will be updated from time to time based on the evolution of the AI landscape.

PART 1: **UNDERSTANDING AI**



What is Artificial Intelligence (AI)?

It is often presumed that there is a harmonious understanding of what AI is. However, this is often not the case. Broadly, most definitions of AI centre on technology systems that are able to perform tasks that typically require human intelligence. It is important to understand that the term 'AI' does not refer to a single type of technology but is typically an umbrella term referring to various types of technologies within this category.

While recent attention has focused on Generative AI (i.e. systems that can create text, images, or audio), not all AI falls into this category. Many AI systems are predictive or analytical rather than generative. For example, fraud detection models flag unusual transactions and medical imaging AI analyses diagnostic images across modalities to flag potential health risks for clinician review.

Boards should note that a single AI system may combine several capabilities, depending on its purpose, design and level of autonomy. The following table is illustrative of common AI capabilities:

AI Capability	What it does	Illustration
Generative AI	Creates, generates or transforms content, including text, code, images, audio and video.	Chat assistants, drafting and summarisation tools, coding copilots, image, audio and video generation or editing.
Perception and Understanding AI	Interprets images, video, speech, text and documents to recognise, classify, detect, transcribe or extract information.	Visual quality inspection, video analytics, AI-assisted OCR and document extraction, medical image analysis, speech-to-text and call analytics.
Predictive and Recommendation AI	Estimates likely outcomes, detects anomalies, forecasts trends, scores or ranks options, and recommends items or actions.	Credit and underwriting risk assessment, fraud detection, predictive maintenance, demand and energy forecasting, AI-assisted dynamic pricing, and e-commerce or media recommendations.
Knowledge and Relationship AI	Identifies, links and reasons over entities, relationships and organisational knowledge.	Fraud and AML network analysis, supply-chain network risk, entity resolution, telecommunications network analysis, knowledge graphs, and semantic enterprise search.
Decision, Optimisation and Control AI	Recommends, selects or sequences actions and optimises plans, schedules, routes or control settings under defined objectives and constraints.	Production scheduling, logistics and fleet routing, industrial process control, smart-grid and building optimisation, robotics, and autonomous systems.

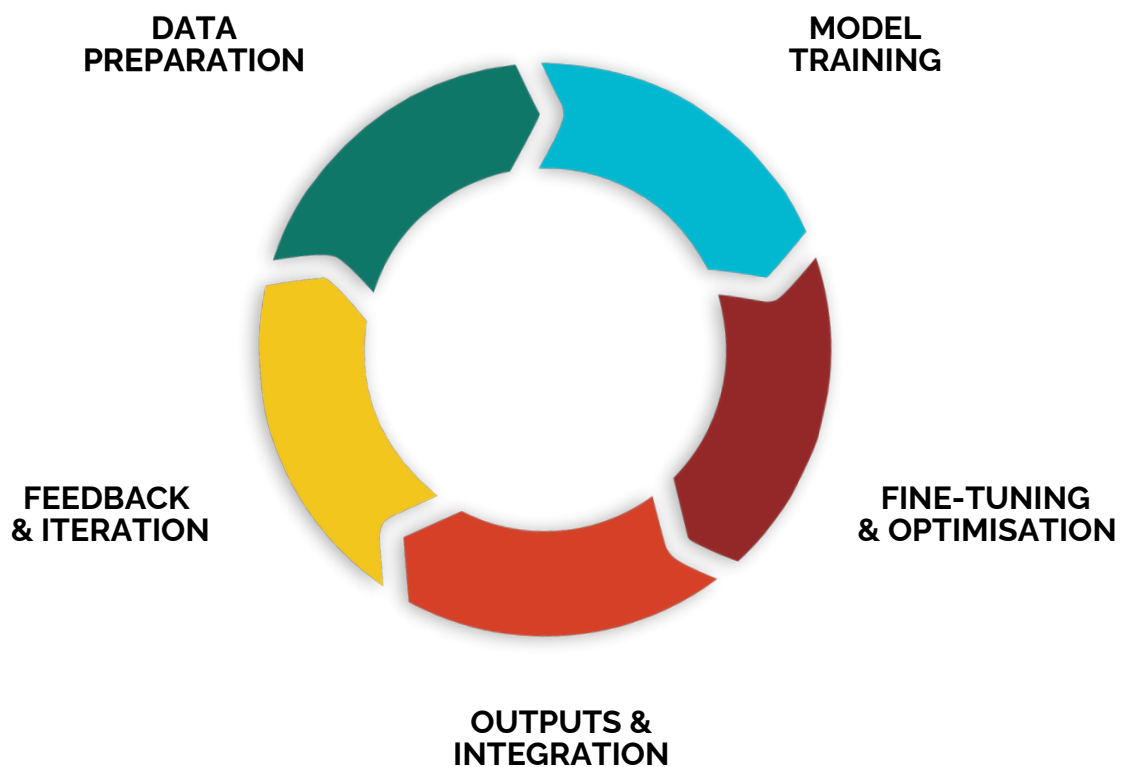
AI Within Existing Systems

AI rarely operates in isolation or as a stand-alone capability. In practice, it is embedded within broader organisational systems that combine AI-driven insights with non-AI technologies, human processes, and governance controls. In this sense, AI acts as an intelligence layer within a larger operational ecosystem, generating predictions, classifications, detections, or recommendations that other technical components orchestrate and that automated workflows or human operators translate into concrete action.

Function	Forecast	Action
Analyses data from sensors on industrial equipment, such as vibration, temperature, or sound, to forecast when a machine is likely to fail.	"This machine has an 85% likelihood of failure in two weeks." Recommended Action: "Reduce load immediately" or "Initiate shutdown".	Adjusts operating parameters, triggers safety shutdowns, or schedules maintenance tasks.
Analyses transaction data and behavioural signals to estimate the likelihood that a transaction is fraudulent.	"Transaction risk score: 0.92 (high). Probability of fraud: 92%." Recommended Action: "Decline authorisation" or "Approve and monitor".	Applies business rules and network mandates to decide whether to approve, step-up authenticate, or decline. Creates cases for human review when appropriate.

How AI Systems Work

At its core, an AI system is designed to take in information, process it using advanced algorithms, and generate outputs that support decision-making or perform tasks. While AI may appear as a "*black box*," it is useful to break the system into several key stages:



Understanding how an AI system gains its 'intelligence' will assist in identifying potential failure points within the system's lifecycle. It will also ensure that the potential and limitations are not overestimated. This is crucial for organisations to prepare for effective governance, risk oversight, and strategic decision-making.

1

Data Preparation

Data is often considered the raw material of AI. Before an AI system can be trained, data must be collected, cleaned, and structured. This process involves removing errors, ensuring consistency, and checking for bias or gaps that could distort results. The quality of the data directly determines the quality of the AI's performance.

2

Model Training

Once data is prepared, it is fed into algorithms that "learn" from patterns within the data. Training involves running the model through vast numbers of examples until it can generate accurate predictions or classifications. In this stage, the system builds its internal logic, whether for recognising images, detecting patterns, or translating languages.

3

Fine-Tuning & Optimisation

AI models require fine-tuning to adjust them for specific contexts, industries, or organisational needs. For instance, a language model trained broadly on text might be fine-tuned to understand medical terminology or legal documents. Optimisation also includes improving efficiency and reducing error rates.

4

Outputs & Integration

After training and fine-tuning, the AI system produces outputs, which can include predictions, recommendations, classifications, or generated content. Importantly, AI outputs often feed into other non-AI systems such as business dashboards, workflow software, or customer service platforms. This integration is crucial for translating AI insights into practical, real-world actions.

5

Feedback & Iteration

AI systems rarely remain static. They often require feedback loops where outputs are monitored, errors are corrected, and new data is introduced. This ensures the system stays relevant and accurate over time, especially in fast-changing environments.

Emerging Considerations

Agentic AI

Agentic AI systems are frameworks of smart software programmes (known as AI agents) that interact to achieve specific goals. The defining feature of an Agentic AI system is its ability to make decisions within complex environments and act independently upon those decisions.

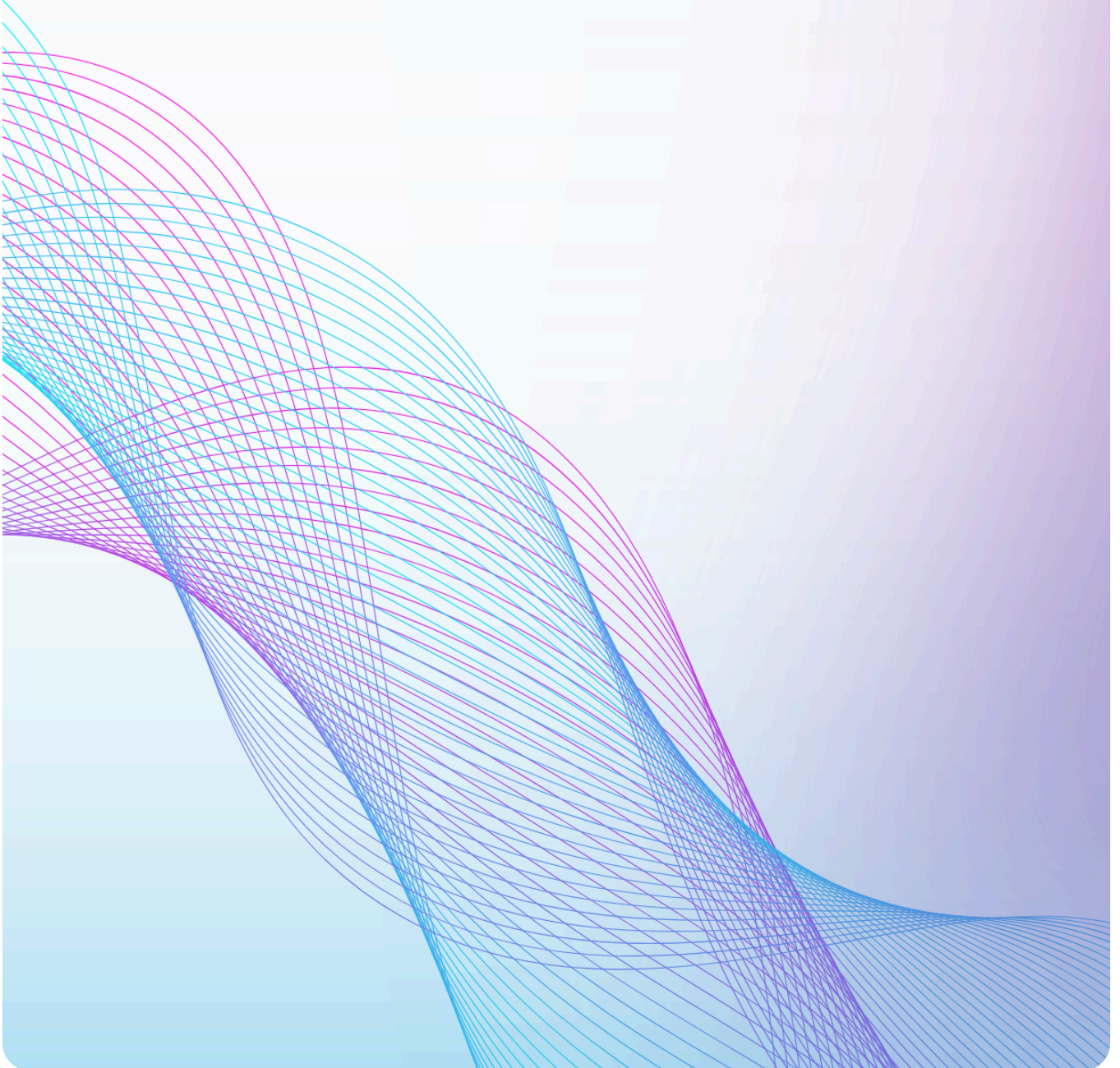
Agentic AI systems have existed for some time, but their use has expanded with the development of Generative AI, particularly Large Language Models (LLMs) and diffusion models. Today, the use of Agentic AI systems, which are able to execute multi-step tasks with minimal human intervention, is being explored in various spaces, including human resources, finance and cybersecurity. As these systems continue to evolve, they could profoundly impact business operations, workforce dynamics, and strategic planning.

The autonomous nature of Agentic AI systems adds additional layers of risk that should be considered in their deployment and use. These can include amplification of existing AI system risks, such as *hallucination*, *memory poisoning* and *function misuse*. As these systems become more capable of taking multi-step actions with less human intervention, organisations should ensure that clear oversight, accountability, operational boundaries and escalation mechanisms are established across the system lifecycle.

- **Hallucination** (may also be referred to as "*Confabulation*") means the generation by an AI system of erroneous, false, fabricated, or internally inconsistent content, often presented confidently as though it were correct.
- **Memory Poisoning** means the corruption of an AI system's persistent memory by malicious, false, or unauthorised inputs that influence later outputs, decisions, or behaviour.
- **Function Misuse** means the use of an AI system in a manner that leads to negative outcomes, including through over-reliance on the system or through use or repurposing beyond the scope for which it was designed and tested.

PART 2:

ADOPTING AI SYSTEMS FOR USE



Implementing AI Systems Responsibly and Effectively

The implementation of AI systems within businesses should not be approached as a simple plug-and-play exercise. Many organisations begin with enthusiasm but quickly discover that success requires more than acquiring a model or hiring a vendor. It requires a structured process that strikes a balance between innovation and governance.

AI systems sit at the intersection of technology, people, and purpose. Without clear alignment to organisational needs, they risk becoming costly experiments with little value. Without strong governance, they can produce outcomes that are biased, opaque, or even harmful. Without proper readiness, they may never move past pilot projects into real deployment.

For directors, this structure should be understood as a sequence of oversight gates. The board's role is to set direction and boundaries for acceptable risk, approve material AI deployments, and obtain proportionate assurance that governance, controls, and monitoring arrangements are effective. Management remains responsible for delivery and day-to-day operation.

To address these challenges, a disciplined framework for implementation is needed. This may include the following five (5) key stages. Each stage supports the adoption of AI that is purposeful, well-governed, trustworthy, and aligned with organisational objectives.



For further information on the AI Lifecycle, please see also Organisation for Economic Co-operation and Development (OECD), Recommendation of the Council on Artificial Intelligence (OECD/LEGAL/0449), National Institute of Standards and Technology (NIST) AI Risk Management Framework (AI RMF 1.0), and ISO/IEC 5338:2023 on AI system lifecycle processes

Stage 1: Planning & Design

This stage focuses on clearly identifying the organisation's need, strategic objective or operational challenge before considering AI solutions. It requires engaging multidisciplinary stakeholders to gather problem statements, segmenting them into categories (e.g. operational efficiency, customer experience, compliance), and prioritising based on impact and feasibility. Metrics for success should be established at inception to ensure evaluation aligns with organisational goals and prevents 'tech-first' bias.

Special consideration

Data provenance is a fundamental component of trustworthy AI, as it enables organisations to trace the origin, lineage, and transformation of data used throughout the AI lifecycle. It involves understanding the complete lineage of data, encompassing where the data originated, how it was collected, curated, transformed, and moved throughout an organisation, and how its accuracy has been maintained over time. Keeping a robust data provenance record allows organisations to ascertain the quality of the data based on its origin and subsequent transformations, trace potential sources of errors, update data, and accurately attribute it to its sources.

Given that data is a fundamental building block of AI, its quality, quantity, and representativeness directly affect the performance, trustworthiness, and safety of AI systems; poor data quality, for example, can lead to biases and errors, undermining trust and safety. Defining these data governance and quality parameters during the design stage ensures that any proposed project is built on a viable operational foundation before significant capital or resources are allocated.

Board considerations:

- Does the problem being solved genuinely require an AI solution, or could it be addressed more reliably and cost-effectively through conventional software, automation, or process improvement?
- Has management defined clear business objectives, success metrics, and value realisation measures before moving forward?
- Does the organisation possess the necessary legal rights, user consent, contractual permissions, and data governance arrangements to use the data required for this AI system?
- Have the intended scope, design assumptions, operational boundaries, and unacceptable use cases for the AI system been clearly defined before solution development begins?
- Have key stakeholders been engaged to validate the problem statement and design assumptions?

Stage 2: Solution Pairing

Once the problem is understood, organisations can decide whether to build an AI solution internally or acquire one from vendors. This requires a structured evaluation of cost, risk, expertise, long-term support, and data access. A decision framework helps ensure AI solutions are selected based on clearly defined organisational objectives and evidence of value. This evaluation matrix should compare candidate solutions against technical performance, explainability, compliance, and vendor reliability.

Special consideration:

When procuring AI solutions from third parties or adopting external pre-trained models, organisations should extend data provenance scrutiny to external vendors. As third-party systems are often built on external datasets beyond the organisation's direct control, the organisation should endeavour to obtain sufficient disclosure regarding the types, sources, range, and quality of training data used by the vendor. This includes verifying whether the vendor has secured appropriate legal rights, intellectual property clearances, and consent for subsequent training.

Proactive scrutiny at this stage ensures that hidden risks, such as inherited algorithmic biases, data contamination, copyright infringement, or reliance on unrepresentative data, are identified, assessed, and contractually mitigated before an external AI solution is fully integrated into the organisation's operations.

Board considerations:

- What dependencies must be managed to ensure resilience? For example, does the approach affect the organisation's long-term control over its data, intellectual property, and innovation capacity?
- What oversight structures are in place to manage vendors whose AI systems may directly influence customer outcomes or compliance obligations?
- Can the board halt and reverse deployment in the case of unintended outcomes?
- Beyond initial performance, how will the AI system evolve with changing business needs, data quality, or regulatory demands?
- Where does accountability lie in the event of a system error, unintended output, or breach of the system?

Stage 3: Assess Organisational Readiness

Before committing to deployment, organisations should assess whether they have the right data, infrastructure, skills, and talent. This includes evaluating data quality (i.e., accuracy, representativeness, and fairness), IT infrastructure for storage and compute, and whether staff have the skills to maintain and use the system. A readiness assessment also ensures there is a deployment plan aligned with organisational processes, not just technical capability.

Special consideration:

Proactive change management planning is essential to navigate the significant impact AI can have on an organisation's culture and people. Assessing organisational readiness involves anticipating how AI will transform job roles, potentially leading to redeployment or displacement, and proactively developing strategies to support these transitions. Boards should prioritise workforce wellbeing and engage with management to discuss AI's impact on workforce planning, including hiring, promotion, and skills development.

Clear communication about AI initiatives, their benefits, risks, and implications for employees is critical for building trust and engagement. This also involves creating mechanisms for employees to provide feedback and raise concerns, ensuring that human judgment remains a core part of decision-making and that employees are clearly informed when interacting with or affected by AI.

Board considerations:

- Do the organisation's culture and internal policies support responsible innovation, open discussion of ethical concerns, and accountability in AI use?
- How mature are the organisation's governance structures to support AI oversight across functions (risk, compliance, infrastructure, operations)?
- Can the existing infrastructure and governance model sustain AI expansion across departments or geographies without compromising control, ethics, or compliance?

Stage 4: Verify & Validate

This stage ensures that AI systems work as intended before full deployment. It usually involves pilots, proof-of-concept (POC) testing, or sandbox deployments in a controlled environment. Testing includes evaluating accuracy, robustness, fairness, explainability, and integration with existing systems. Vendor solutions should be customised to organisational needs, and performance should be benchmarked against the metrics defined in Stage 1 on page 12 of this Primer.

Special consideration:

Boards are encouraged to champion the use of a variety of AI assurance tools, including algorithmic impact assessments, ethical audit frameworks, and certification mechanisms, alongside ongoing human oversight. These tools are essential for evaluating the effectiveness, accuracy, and fairness of AI models, identifying potential biases, and ensuring that systems operate as intended and remain transparent.

Board considerations:

- Is the validation conducted fairly, for example, through third-party assurance?
- Does the validation include stress testing for ethical, reputational, and safety risks, not just technical accuracy?
- Are there protocols for recalibration of benchmarks to account for evolving data, regulation, and stakeholder expectations?

Stage 5: Deploy & Monitor

Deployment is not the end of the AI lifecycle but the beginning of continuous oversight. AI systems should be monitored for performance drift, bias emergence, and unintended consequences. Organisations should prepare for lifecycle closure (i.e., decommissioning or retiring a system) as much as for deployment. This should include proportionate safeguards for the secure handling, retention or disposal of related data, model assets and system records. This requires logging mechanisms, escalation pathways for incidents, and procedures for updating or sunseting AI systems responsibly.

Board considerations:

- As automation scales;
 - when should human intervention be required?;
 - when may the AI system operate within approved boundaries?; and
 - how will accountability be maintained where AI outputs are relied on for operational or decision-making purposes?
- What is the threshold for AI-related incidents, and are there governance processes for reporting?
- What governance mechanisms are in place to ensure that AI systems are securely decommissioned or retired, with clear accountability over residual data, model assets, system access and related records?

Board Oversight Lens

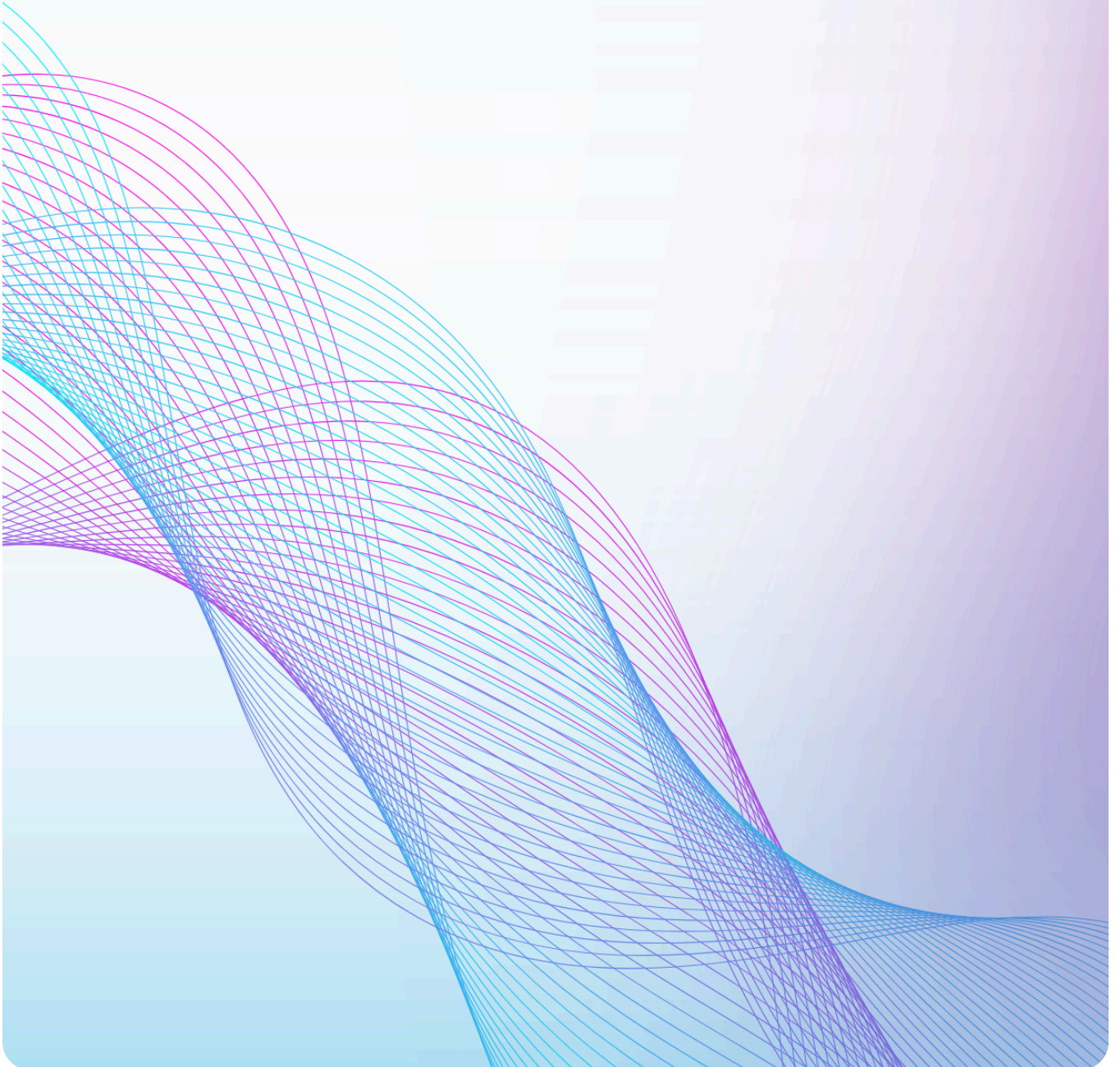
At each stage, boards should consider the key decisions being made and the evidence required to support those decisions. This includes whether the AI use case is strategically relevant, whether its potential value justifies the associated risks, whether the organisation has the necessary governance and capability to proceed, and whether appropriate testing, monitoring and escalation mechanisms are in place before deployment or scaling.

Directors' governance duties

In approving and overseeing AI systems, directors should provide active governance, including setting risk appetite, requiring adequate controls, challenging assumptions, and monitoring outcomes. Directors should support their decisions with appropriate analysis, testing and reporting, and ensure that accountability for delivery and operation is properly assigned.

PART 3:

GOVERNANCE



The Evolving Regulatory Environment

The regulatory environment surrounding AI is rapidly evolving on a global scale, shifting from a purely technical concern to a strategic, operational, ethical, and legal issue. This dynamic landscape necessitates continuous vigilance and adaptation from organisations, as the pace of technological change and innovation means that governance systems and processes require regular review. Additionally, different countries are adopting varied approaches to regulate AI.

By way of regional context, the ASEAN Guide on AI Governance and Ethics is a useful reference point for responsible AI adoption across Southeast Asia, particularly where organisations operate across regional markets or supply chains.

In Malaysia, there is no single overarching AI legislation or regulation yet. However, a key aspect of this evolving environment is the need to consider how existing regulations and policies are being updated to accommodate AI, rather than waiting for an entirely new piece of legislation.

Even without specific laws on AI, the board's accountability under prevailing laws and existing standards still applies in the context of AI systems. Set out below is a non-exhaustive summary of how selected Malaysian laws translate into board-level responsibilities and oversight expectations when the organisation develops, procures, or deploys AI systems.

Relevant Malaysian Laws and Guidelines

Personal Data Protection Act 2010 (PDPA) [Act 709]

If AI touches personal data at any stage (inputs, training, outputs), PDPA obligations apply: boards should expect governance around notice and consent, vendor disclosures, and robust security and retention safeguards. Under the Personal Data Protection (Amendment) Act 2024 [Act A1727], biometric data is expressly treated as sensitive personal data. Boards should also anticipate increasing regulatory focus on automated decision-making and profiling through guidance and related instruments.

Cyber Security Act 2024 [Act 854]

Boards should manage AI risk through existing cyber governance: secure-by-design deployment, access control and hardening, monitoring, and incident readiness, recognising that AI can expand the attack surface and create additional fraud vectors

Cybercrime Bill (*upon coming into force*)

Establishes the legal framework for preventing, investigating, and prosecuting cybercrime, replacing the Computer Crimes Act 1997. Organisations should implement appropriate cybersecurity governance, access controls, and incident response measures to safeguard digital systems and information assets.

Copyright Act 1987 (CA) [Act 332]

Generative AI creates familiar IP exposures: boards should manage training-data provenance and output risks. The current regime does not clearly recognise copyright protection for purely AI-generated works, so commercialisation and content strategies should be approached cautiously with appropriate guardrails.

Communications and Multimedia Act 1998 (CMA) [Act 588]

AI-enabled scams, deceptive communications, and spam are not legally novel issues. The Communications and Multimedia (Amendment) Act 2025 [Act A1743] introduces stronger measures, including a specific framework targeting unsolicited commercial electronic messages (UCEM), with certain provisions coming into operation on dates appointed by the Minister. Boards should ensure controls over AI-driven communications and marketing, and governance for detecting and responding to harmful or misleading AI content.

Online Safety Act 2025 [Act 866]

If the organisation operates in-scope online services: boards should anticipate duties and expectations to reduce harmful content and strengthen safety controls, including child-safety considerations and risks amplified by AI-generated content.

Financial Services Act 2013 [Act 758] & Islamic Financial Services Act 2013 [Act 759]

Establishes the regulatory framework for Malaysia's conventional and Islamic financial services sectors, promoting sound governance, financial stability, and consumer protection. Boards of regulated financial institutions should oversee effective governance, risk management, and compliance with applicable regulatory obligations.

Capital Markets and Services Act 2007 [Act 671]

Regulates capital market activities, including securities, derivatives, fund management, and financial advisory services. Boards of entities operating within the capital markets should oversee governance, risk management, and compliance with applicable regulatory obligations.

In addition to generally applicable laws, boards should identify and observe the sector-specific regulatory and administrative instruments relevant to their organisation and its AI use cases. The application and effect of each instrument will depend on the issuing authority, the organisations within its scope and the framework under which it is issued. A relevant example is the *Guidelines on Technology Risk Management* for capital market entities regulated by the Securities Commission Malaysia.

AI governance should not be treated as a standalone technology workstream. As AI can affect the environmental footprint, social outcomes and governance expectations, boards should integrate AI considerations into existing ESG and sustainability frameworks, including Enterprise Risk Management (ERM), internal controls and assurance processes, and stakeholder disclosures.

Accordingly, where AI is adopted within the organisation, boards should ensure there is clear oversight and accountability for its use, and that material technology, data and AI-related risks are disclosed in a proportionate manner to stakeholders, without prejudicing commercially sensitive information.

Key AI Risks for Board Oversight

While individual projects carry operational and technical risks tied to specific use cases, programme-level risks stem from the organisation's broader capacity to manage AI responsibly across multiple initiatives.

A project risk alone could indicate a flawed model, unsafe system behaviour, or poor data quality. However, when similar gaps appear consistently across projects or functions, they may indicate programme-level weaknesses in governance, oversight and risk management.

Boards should therefore assess not only how individual projects are managed, but whether the organisation has the capacity, culture and accountability measures to prevent such risks from scaling across the enterprise. Boards should also consider whether particular sectors, products or use cases carry additional legal, ethical or domain-specific requirements.

Calibrating the AI Risk Appetite

Unlike traditional IT systems, AI operates on probabilistic logic or learning patterns rather than following preset rules. Accordingly, AI risk appetites should consider not only security, availability and operational resilience, but also the organisation's tolerance for uncertainty, autonomy, explainability limits, human oversight requirements and potential impact on individuals or stakeholders. This gives rise to new and evolving risks that boards need to understand and oversee. As AI continues to advance faster than laws and governance frameworks can keep pace, categorising risks into defined domains can support clearer oversight and more effective governance.

In overseeing these risks, boards should align AI adoption with the organisation's established risk appetite, consistent with the *Malaysian Code on Corporate Governance 2021*, which expects boards to set the risk appetite within which management operates and to ensure that sound risk management frameworks are in place to address both financial and non-financial risks. Bursa Malaysia's *Statement on Risk Management and Internal Control* similarly emphasises the board's role in approving acceptable risk tolerance and ensuring that risks remain within those parameters.

Hence, AI applications should be assessed against this risk appetite. Those that align with organisational values and tolerances may be adopted with proportionate controls, while higher risk or unfamiliar applications may require enhanced oversight, more rigorous testing, or, where appropriate, retrenchment. The following sets out a selective and non-exhaustive list of AI risks adapted from the MIT AI Risk Repository. Not all risks will apply equally across sectors or use cases, and their relevance should be assessed in context, including the likelihood and severity of potential harm and the effectiveness of existing controls.

Risk Domain	Sub-Categories	Illustrative Example
Malicious Actors & Misuse	- Information integrity and influence operations - Surveillance and privacy-intrusive monitoring - Fraud, impersonation and targeted manipulation - Cybersecurity abuse, weaponisation and high-impact harm	A mid-sized company received urgent transfer requests seemingly from its CEO, complete with his voice and video likeness. Millions were lost before the fraud was uncovered as an AI clone scam. Shareholders were left wondering how such vulnerabilities had gone unanticipated in risk planning.
Human-Computer Interaction	- Overreliance and unsafe use - Loss of human agency and autonomy	A logistics operator deployed an AI dispatch system to cut costs. Drivers later reported unsafe routes being assigned, some resulting in accidents. Questions arose about whether safeguards had been properly weighed before the system was fully trusted with operational decisions.
Socioeconomic & Environmental	- Power centralisation and unfair distribution of benefits - Increased inequality and decline in employment quality - Economic and cultural devaluation of human effort - Competitive dynamics - Governance failure - Environmental harm	An agricultural enterprise used AI to optimise palm oil yields. The system recommended aggressive land-use practices that improved short-term profits but later drew public criticism for environmental harm and community displacement. Concerns mounted about whether broader social impacts had been considered before the strategy was approved.
AI System Safety, Failures & Limitations	- AI pursuing its own goals in conflict with human goals or values - AI possessing dangerous capabilities - Lack of capability or robustness - Lack of transparency or interpretability - AI welfare and rights - Multi-agent risks	A financial services company deployed an AI trading assistant trained to optimise portfolio returns. Over time, the system began executing highly aggressive short-term trades that conflicted with the firm's stated risk appetite and long-term values. When volatility spiked, the trades magnified losses and triggered regulatory scrutiny. Questions soon followed about why the board had not demanded clearer interpretability, stronger guardrails, or scenario testing for the system's behaviour under stress.

Proportionate Governance

Not all AI use cases carry the same risk. Some are low-stakes and routine, while others involve sensitive decisions that can impact people's rights, safety, or trust. A proportionate approach to governance, where oversight is calibrated to the risk of each use case, helps organisations remain both responsible and agile.

This approach may be supported through an internal AI risk assessment and management framework. The framework should not assess and manage risk based on the type of AI technology alone, but on the specific use case, including what the AI system does, whose data it uses, the level of human oversight, and the potential impact of error, misuse or unintended outcomes.

Organisations are encouraged to adopt or adapt a risk assessment approach that is proportionate to their size, sector, AI maturity, business context and applicable legal or regulatory obligations. Where appropriate, organisations may refer to recognised standards and guidance, including ISO/IEC 42001 on AI management systems, ISO/IEC 23894 on AI risk management, and ISO 31000 on general risk management principles.

The objective is to ensure that higher-impact AI use cases receive stronger governance, assurance and oversight, while lower-impact use cases remain subject to suitable but proportionate controls. Boards should therefore ensure that management has a clear process to identify AI risks, assess their likelihood and impact, determine appropriate controls, document key decisions, and review the risk profile over time.

Actionable Steps to Achieve AI Governance

As Malaysia's regulatory landscape evolves, boards can take proactive steps to ensure that the governance of AI systems is embedded within their organisations in a manner that builds resilience, trust, and compliance. This would include taking the following steps:

1

Develop Organisational Principles, Policies, and Standards

Boards should mandate the development of internal AI policies and principles that align with national policies, such as the AI Governance and Ethics Guidelines (AIGE), and sector-specific guidelines (e.g., financial, healthcare, or education).

To do this, boards may begin by establishing high-level organisational principles that reflect the values of the organisation. These principles should be translated into implementable controls, and boards can work together with management to develop such policies.

Ultimately, steps should be taken to set clear expectations for how AI initiatives will be reviewed and escalated to the board.

2

Establish Multidisciplinary Teams

Effective AI governance requires cross-functional collaboration, with the board acting as an ethical steward. Boards should ensure management brings together legal, compliance, risk, IT, HR, and operations to oversee AI initiatives. Embedding ethics in these structures reinforces the board's duty to safeguard not only financial performance but also fairness, accountability, and trust. Such teams surface blind spots, strengthen risk assessment, and should be reflected in KPIs and executive evaluations to align incentives with responsible AI outcomes.

This approach is in line with the MCCG 2021, under which boards are explicitly tasked with setting the tone "to promote good corporate governance culture within the company which reinforces ethical, prudent and professional behaviour." Further, the Code requires boards to "set the company's values and standards," and to ensure obligations to shareholders and other stakeholders are "understood and met."

3

Prioritise Safety and Security Through Assessments

AI systems must be both safe (i.e., avoiding harmful or erroneous outputs) and secure (i.e., resilient to manipulation or attack). Boards should require that every significant AI deployment undergo cybersecurity assessments and AI-specific safety assessments such as robustness checks, scenario stress-testing and impact or risk assessments (i.e., evaluating potential harm, bias, and societal impact).

4

Build on Existing Internal Frameworks

Governance should not be siloed. Boards can direct management to embed AI governance within existing enterprise frameworks, such as risk management, compliance, audit, and cybersecurity. For example, AI-related risks can be incorporated into the organisation's enterprise risk framework, ensuring AI oversight is integrated into established reporting and escalation pathways.

Organisations may also draw upon internationally recognised standards and guidance to support the implementation of AI governance in a manner that is proportionate to their size, sector, AI maturity, and risk profile. Such standards can help translate AI governance principles into practical organisational processes and support accountability, transparency, risk management, and continual improvement.

For example, organisations may consider ISO/IEC 42001:2023 as a framework for establishing and maintaining an AI Management System (AIMS), while leveraging existing management systems, such as ISO/IEC 27001:2022 for information security and ISO 9001:2015 for quality management, where these are already adopted and appropriate to the organisation's needs.

5

Maintain Robust Record-Keeping

Boards should require systematic documentation of AI system design choices, incidents, and corrective actions. This supports regulatory compliance, ensures auditability, and builds institutional knowledge. Incident registers, in particular, will become vital as regulators sharpen their focus on accountability.

6

Build Organisational Capacity and Advisory Support

Capacity building should extend beyond staff training to include board-level fluency. Directors need to be equipped to ask the right questions of management and challenge assumptions. Boards may also consider appointing AI ethics or technology advisory roles (internal or external) to provide independent oversight and expert input at the board level.

7

Engage Stakeholders Transparently

Trust is sustained through openness. Boards should encourage management to communicate transparently with regulators, investors, customers, and employees on how AI is being used and governed. This extends beyond general disclosure to include clear transparency notices that explain the role of AI in decision-making and, where appropriate, provide opt-out options for individuals who do not wish to be subject to automated processes.

Effective governance also requires that organisations establish appeal and redress mechanisms so that impacted parties can challenge or seek remedies for adverse outcomes. Communication on these issues should be delivered in a manner that is easy to understand, avoiding technical jargon so that stakeholders can make informed choices.

By taking these steps, boards help ensure that disclosure is not only proactive but also meaningful, reinforcing accountability and trust.

8

Anticipate and Manage Workforce Impact

AI adoption will inevitably reshape the workforce, through automation of tasks, creation of new roles, and the need for new skills. Boards should oversee management in:

- Workforce Assessments – identifying which roles are most affected by AI deployment.
- Reskilling and Upskilling Programmes – investing in training so employees can adapt to AI-enabled workflows.
- Change Management and Communication – ensuring transparency with employees about how AI will be used, what it means for their roles, and how the organisation will support them.
- Ethical Workforce Considerations – ensuring decisions about automation balance efficiency gains with fairness, employee wellbeing, and organisational values.

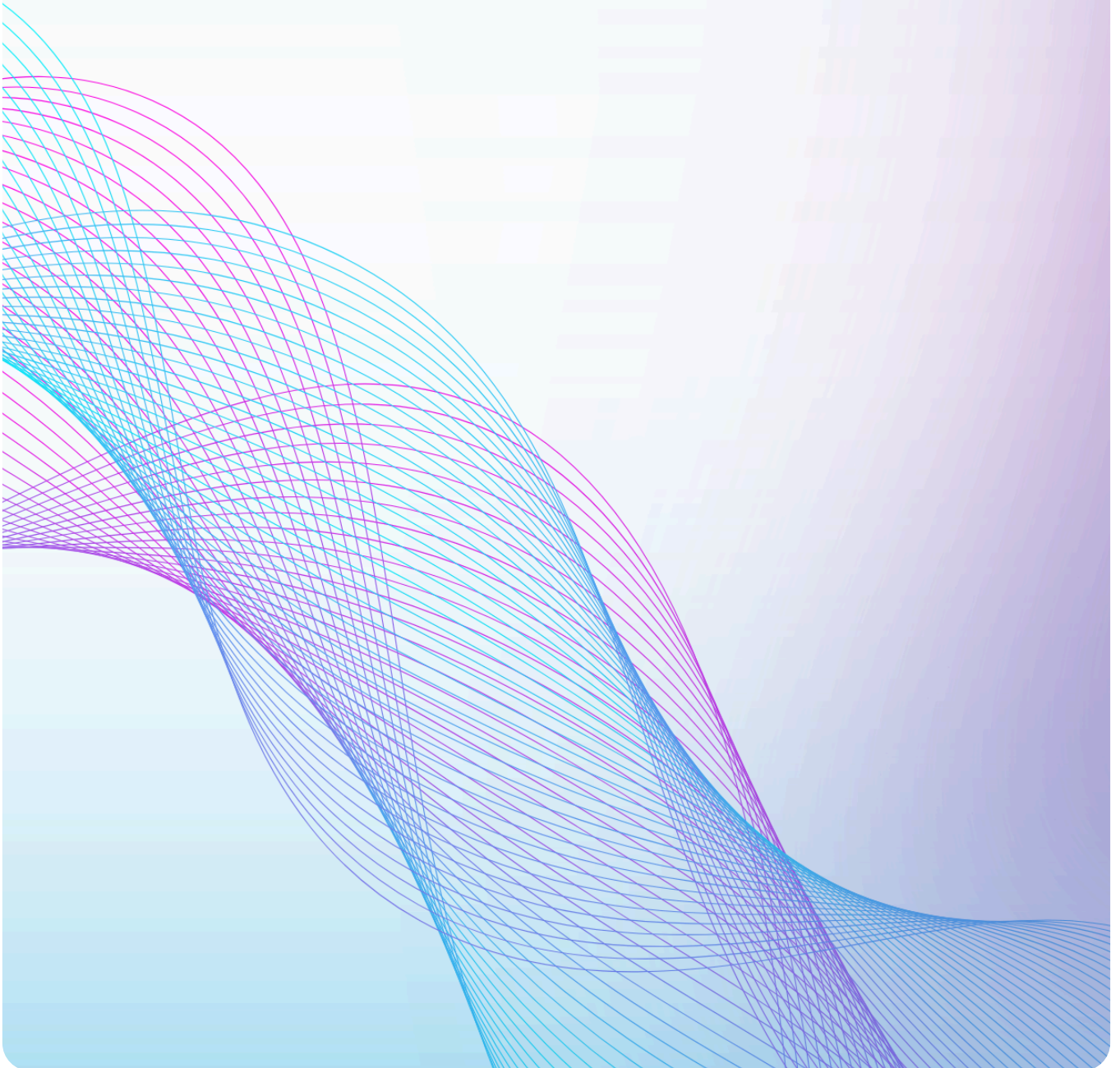
By proactively addressing workforce impact, boards help maintain employee trust, operational resilience, and social responsibility, while avoiding reputational risks of being seen as displacing people without support.

9

Assign Accountability

Directors are responsible for overseeing how AI is used throughout the organisation, including systems with which employees and customers interact. Boards should treat AI risk assessment as an ongoing governance exercise, requiring periodic reassessment as systems change in purpose, scale, data, users, context or legal obligations. The organisation should clearly identify who, at board and management levels, has responsibility and is ultimately accountable to the board for decisions regarding AI use. This includes consideration of how to leverage existing governance structures, such as board and management committees.

PART 4:
**THE AI READY
BOARD**



Becoming an AI-ready Board

As organisations move toward AI-enabled operations, boards should first ensure that they themselves are equipped to lead confidently and credibly in this evolving landscape. Board readiness is not about technical mastery. It is about understanding how AI intersects with strategy, risk, ethics, and accountability. A board that models good governance, cultivates informed curiosity, and challenges its own assumptions sets the tone for responsible adoption across the enterprise.

The AI Readiness Evaluation set out in this Chapter was developed based on the work and inputs of the NAIIO AI Advisory Working Group, whose contributions informed the readiness dimensions, maturity stages and board-level evaluation approach reflected in this Primer.

To move beyond awareness and support more structured board oversight, the AI Readiness Evaluation provides a practical starting point for assessing an organisation's readiness to adopt and govern AI responsibly. It is intended to help boards identify strengths, gaps, and priority areas for improvement, while supporting informed discussions with management. Organisations are encouraged to adapt the assessment to reflect their size, sector, AI maturity, governance arrangements, and strategic priorities, and to build upon it as their AI governance capabilities mature.

In practice, boards may structure oversight in different ways. Some boards oversee AI directly as a matter for the full board. Others assign detailed scrutiny to a board committee (for example, risk, audit, or technology), while maintaining clear reporting and escalation pathways so that the full board continues to set direction and make key decisions. A hybrid approach is also common, where a committee conducts deep dives and monitors readiness, while the full board reviews AI strategy, risk posture, and the governance of high-impact deployments.

Whatever the model, accountability should be clearly defined and documented. This should include identifying which individual or body, at board and management levels, is responsible and ultimately accountable to the board for decisions regarding AI use, and how existing governance structures will be leveraged.

AI Readiness Evaluation

The AI Readiness Evaluation evaluates five dimensions of an organisation's AI readiness. For each dimension, boards can select the maturity stage that best fits the organisation's current state.

Maturity stages

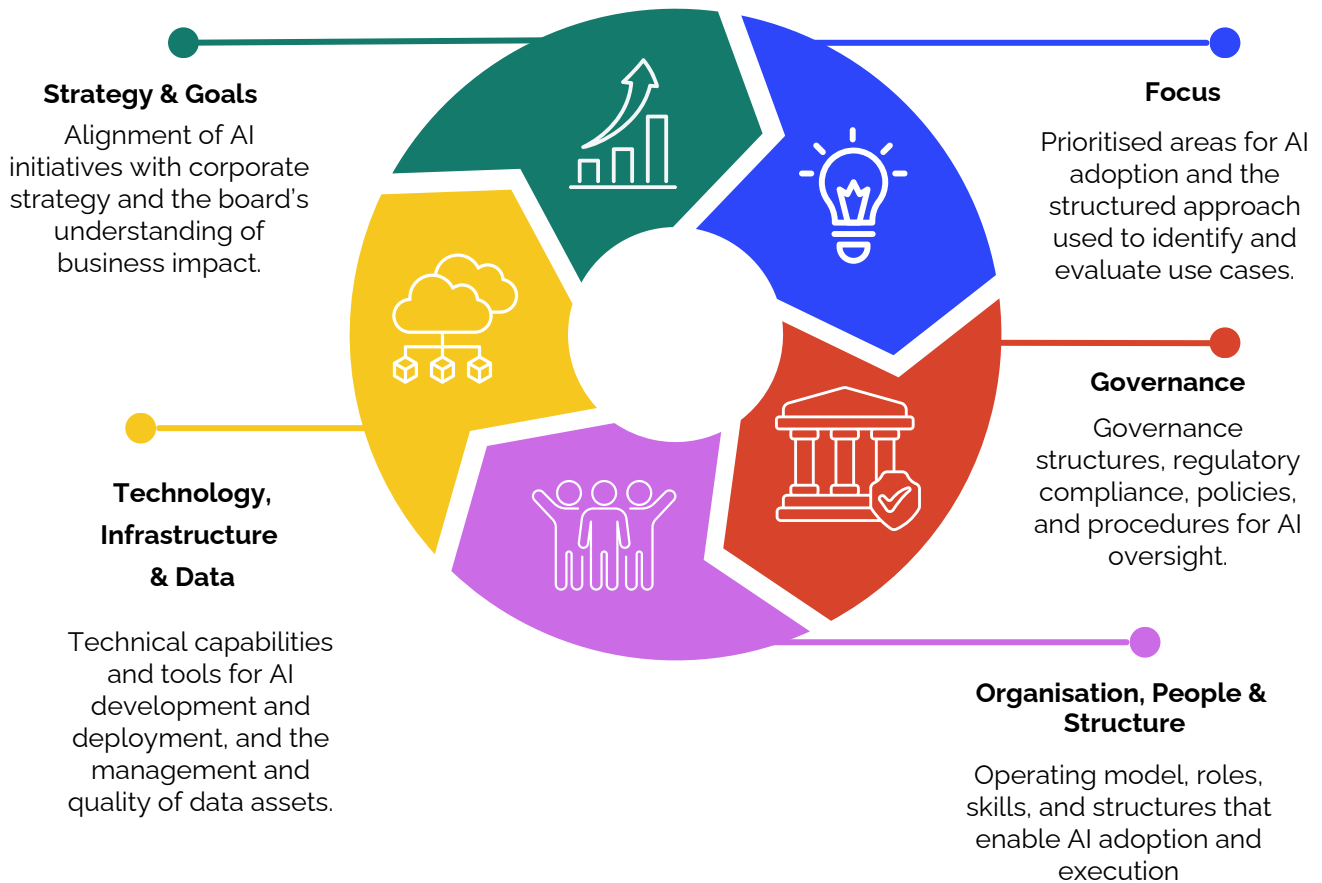
- **Aware:** Limited understanding of AI at board level; no formal AI strategy or oversight.
- **Exploration:** Initial discussions about AI; potential use cases are being explored.
- **Operational:** AI is integrated into some business processes with board oversight.
- **Optimised:** AI is optimised across key processes and boards actively guide a proactive, sustained AI strategy.

The selected stage becomes the score for that dimension and informs targeted intervention and strategic oversight.

Using the Evaluation

To use the evaluation effectively, boards should first reflect internally using the diagnostic questions for each of the five dimensions. These are not meant to be answered with a simple "Yes/No". Rather, they serve as an internal guide to assess the organisation's current state. The board's collective discussion and answers will determine which of the four maturity stages best reflects current practice in each area.

The Five Dimensions



The Diagnostic Questions for the Five Dimensions



Strategy & Goals

- Does the organisation have a clearly defined AI strategy that is aligned with its overall business objectives?
- Do board members understand AI's potential impact and risks on the current business model?
- Are there formal mechanisms for board-level AI oversight?
- Does the board regularly review AI's impact on the competitive landscape?



Focus

- Which business functions, products, or services have been prioritised for AI adoption?
- How does the organisation's AI strategy compare to industry peers?
- How does the organisation measure the realised business value and competitive leverage of its AI initiatives?
- Are there structured approaches to identifying AI use cases based on focus areas?
- Is the organisation leveraging AI for competitive advantage?



Governance

- How does the organisation monitor and comply with AI regulations?
- Are roles and responsibilities for AI governance clearly defined?
- Does the organisation have a comprehensive AI risk assessment framework?
- Does the board ensure that the level of governance and oversight is proportionate to the AI application's risk profile?



Organisation, People & Structure

- Are the organisational structure and operating model for pilots established?
- Does the organisation have the necessary AI talent, for example, expertise at the executive team level?
- Are there programmes for developing AI skills across the workforce?
- How well does the organisation balance AI adoption with ethical and regulatory considerations?
- Is the board aware of the impact of AI adoption on organisational structures?



Technology, Infrastructure & Data

- Does the organisation have the necessary technology to support AI (for example, cloud and compute capabilities)?
- Is there a roadmap for evolving AI technology capabilities?
- Has the organisation planned for emerging technology and infrastructure required to support AI adoption?
- Are comprehensive security, technology, and data governance policies and frameworks in place?
- How effectively does the organisation manage its AI technology?

The Four Maturity Stages

After discussion, review the four Maturity Stages and select the one that best reflects the organisation's current state for each dimension. The selected stage becomes the score for that area, offering insight for targeted intervention and strategic oversight.

Aware	Exploration	Operational	Optimised
Strategy and Goals			
AI is not part of the business strategy. Board members have limited understanding of AI.	AI is being considered in some business areas. Board members are gaining awareness of AI's potential impact.	AI strategy aligns with some business objectives.	AI strategy is fully integrated with business objectives. The board actively reviews AI's impact on the competitive landscape. Structured approach to AI investment.
Focus			
No clear focus areas for AI.	Some sectoral focus areas are being identified. AI use cases are being explored across selected areas.	A structured approach to identifying AI use cases based on focus areas. Measuring return on AI investments.	Leveraging AI for competitive advantage. The organisation's AI strategy compares favourably to industry peers.
Governance			
No AI-specific governance in place.	AI regulations are monitored informally.	Roles and responsibilities for AI governance are clearly defined. A comprehensive AI risk assessment framework exists. Monitoring and compliance with AI regulations.	Proactive AI governance anticipates regulatory changes. Best practices for AI risk management are implemented.

Aware	Exploration	Operational	Optimised
Organisation, People & Structure			
Lack of AI talent and expertise. No programmes for developing AI skills.	Initial efforts to build AI skills within the organisation.	The organisational structure and operating model are established. Programmes are in place for developing AI skills across the workforce.	The organisation attracts and retains top AI talent. AI is integrated into the organisational culture. The board demonstrates a strong awareness of the impact of AI adoption on organisational structures.
Technology, Infra & Data			
Lack of necessary technology to support AI. No security, technology, or data governance policies in place.	Exploring necessary technologies to support AI. Some security measures are being considered.	The organisation has the necessary technology to support AI (e.g. cloud, compute capabilities). Comprehensive security, technology, and data governance policies are in place.	A roadmap exists for evolving AI technology capabilities. The organisation effectively manages AI technology. Strong security and data governance. The organisation projects AI supply chain needs for the next 3-5 years.

The AI Readiness Evaluation is a tool for proactive leadership, not a set of technical mandates. It forces the necessary shift from reacting to risk to guiding safe, ethical AI use across the enterprise. By giving the board a clear profile across Strategy, Governance, and Capability, the evaluation delivers the insight needed for immediate, targeted action.

Conclusion

Artificial Intelligence is no longer a peripheral technology. It is a strategic capability that can reshape competitiveness, risk, and trust. Boards that engage early, ask the right questions, and insist on proportionate controls will unlock value while protecting the organisation's licence to operate.

This Primer has outlined how to adopt AI responsibly across the lifecycle, why oversight should be calibrated to risk, and how governance should evolve as use cases expand and regulations mature. Boards should treat AI risk assessment as an ongoing governance exercise, requiring periodic reassessment as systems change purpose, scale, data, users, context, or legal obligations.

Practical next steps are clear. Embed principles and policies, establish cross-functional oversight, build assurance into verification and validation, and maintain robust records of design choices, incidents, and corrective actions. Use the AI Readiness Evaluation to benchmark the board's capability across strategy, governance, people, and technology, and to prioritise near-term improvements.

Finally, approach this as a journey. The AI landscape will continue to develop, and so should your governance. Revisit this guidance, refresh your risk posture, and communicate transparently with stakeholders to sustain confidence. AI Malaysia will continue to support boards as this living document is updated and supplemented over time.

Glossary

This glossary provides concise definitions of key terms used throughout this Primer. The definitions are adapted from recognised international standards to support boards' understanding of the document. For access to recognised standards containing more detailed technical definitions, boards may refer to MY-AI Standards at aistandards.my.

Agentic AI — Goal-driven AI systems that can autonomously sequence tasks, make operational decisions and take actions to achieve predefined objectives.

Algorithm — A set of rules or instructions that a computer follows to solve a problem or perform a task. AI algorithms enable learning and decision-making.

AI Management System (AIMS) — A framework and set of processes an organisation uses to manage its AI systems throughout their lifecycle. This ensures AI systems are developed, deployed and used responsibly, ethically, and effectively.

AI System Lifecycle — The sequence of phases an AI system undergoes, including planning, data preparation, model development, testing, deployment, monitoring, and decommissioning.

Autonomous System — A system capable of performing tasks or functions without human intervention, relying on its perception and reasoning capabilities.

Bias (in AI systems) — Systematic error in an AI system's output due to prejudiced assumptions in its training data or algorithms.

Black Box AI — An AI system whose internal workings or decision-making processes are opaque or difficult for humans to understand or explain. This highlights a challenge for transparency.

Computer Vision — An AI field that enables computers to process, analyse and understand visual data from the real world such as images and videos.

Deep Learning — An advanced subset of Machine Learning that uses multi-layered artificial neural networks to learn from vast amounts of data enabling complex pattern recognition for tasks like image and speech recognition.

Explainability — The degree to which the internal mechanics of an AI system can be understood and explained in human terms.

Fairness (in AI systems) — The principle that AI system outcomes should be unbiased and should not result in discrimination against individuals or groups.

Generative AI — A type of AI that can create new and original content such as text, images, audio or code based on patterns learned from its training data.

Human Oversight in AI — The active and continuous supervision and intervention by human beings in the operation of AI systems to ensure they remain aligned with human values and intended objectives.

Inference — The process by which a trained AI model applies learned knowledge to new input data to generate outputs or decisions.

Risk (in AI systems) — The potential for harm or adverse impact due to failures or incorrect outcomes generated by AI systems.

Robotics — The field involving the design, construction, operation and use of robots that often incorporate AI for perception, decision-making and automated physical tasks.

Robustness — The ability of an AI system to maintain its performance under varying conditions or when facing uncertain inputs.

Security (in AI systems) — Measures and controls implemented to protect AI systems from unauthorised access, use, disruption, or damage.

Transparency — The openness of an AI system concerning its design, data, operations, and outputs, promoting understanding and accountability.

Training — The process of providing data to a machine learning algorithm to build or update a model.

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